



German International University
Faculty of Informatics and Computer Science

Axis - Intelligent Posture and Alignment Detection

Bachelor Thesis

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TA Mayar Soliman

Submission Date: 5 January, 2026



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This is to certify that:

- (i) The thesis comprises only my original work toward the Bachelor Degree
- (ii) Due acknowledgement has been made in the text to all other material used
- (iii) Artificial intelligence tools were used exclusively for language refinement and presentation purposes. No AI tools were used to generate research ideas, methodology, results, analysis, or conclusions. The author remains fully responsible for all content of this thesis.

Jasmine Mohamed Hany Maher
5 January, 2026

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I would also like to take this opportunity to thank all those individuals, directly or indirectly involved, for successfully carrying out this graduation project. First and foremost, I would like to express my gratitude to all project supervisors: Dr. Nada Sharaf, Dr. Slim Abdennadher, Dr. Mahmoud Ibrahim, Dr. Amal Yassin, and TA Mayar Soliman for their loyal support and assistance regarding this project and for developing this project, as well as upgrading my knowledge and awareness level towards computer science and their diverse applications. Thank you from the bottom of my heart to each of them.

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Chapter 1

Abstract

People often have posture which can cause problems with the muscles and bones and lead to back and shoulder issues and pain that will not go away. The usual ways to check posture are not very good because they take a lot of time can be painful or need an expert to look at you. For example someone has to watch you or take pictures of your body. So doctors really need a way to check posture that is easy does not hurt and gives them the information they need to help people get better. This new posture analysis system can help doctors make decisions about what is wrong, with the patient and how to fix it.

This study proposes a computer vision-based system for analyzing patient posture using front and rear two-dimensional (2D) images. The system leverages the MediaPipe Pose framework, which detects 33 anatomical keypoints, along with additional posture-specific keypoints including C7, hip center, torso center, and two points labeled A and B. Images undergo preprocessing steps such as background subtraction, normalization, cropping, and reflection to enhance landmark detection accuracy. Keypoints are annotated through a combination of automated detection and manual verification to ensure reliability.

Posture deviation is quantified by calculating the angle formed between the torso reference points, allowing assessment of leaning and misalignment. The system is evaluated using the Percentage of Correct Keypoints (PCK) metric, comparing predicted landmarks against manually annotated ground truth. Results demonstrated high accuracy, with PCK exceeding 0.92 for shoulder points and approximately 0.88 for the torso center. In addition, the system generates visual reports and dashboards to track postural changes over time and provide personalized posture improvement recommendations.

Overall, this research validates that 2D pose estimation combined with geometric analysis is an effective, non-intrusive clinical posture evaluation method. It facilitates early diagnosis, treatment planning, and rehabilitation interventions while minimizing manual assessment efforts.

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Chapter 2

Introduction

2.1 Motivation and Problem Statement

Postural misalignment is a common health issue that affects all ages. Cases like scoliosis, shoulder alignment problems, and torso misalignment are known to severely affect one's quality of life due to musculoskeletal pain and loss of mobility and functionality in the long run [1, 2]. The early diagnosis and tracking of the misalignment of posture are henceforth vital to ensure that the issue does not worsen and to utilize proper and efficient therapy treatment. Yet conventional systems available to diagnose the postural alignment issue have some important shortcomings. In traditional diagnostic methods using radiographic images, the diagnostic process can accurately gauge precise information. However, this method is associated with radiation exposure and is not feasible to use frequently [3]. Postural alignment diagnosis via expert professionals assessments/physiotherapists takes a long time and is subjective and inconsistent among different observers [4]. Challenges associated with these diagnostic methods emphasize the demand for radiation-free and efficient solutions to conduct the process of posture analysis.

Posture analysis has been found to be an area which has not received adequate attention as yet, especially as it pertains to computer vision-based systems [5]. This is surprising, especially since there has been an increase in the incidence of muscular problems such as scoliosis, shoulder deformity, and trunk deviations [1]. During the course of research, it has been found that there has been limited work carried out as yet, which has been related to this project [6]. The dearth of studies on this issue brings to the fore the large gap that exists in this field, where traditional techniques for the assessment of posture have been well established but, in most cases, are prone to observation either manually or with the help of imaging techniques [3].

With these problems, it can be detected that there occurs the need for developing techniques to enable an accurate, efficient, and non-invasive posture analysis system [7]. The motivation to work on this topic has been developed to contribute to helping healthcare experts in developing an accurate system to aid in the analysis, diagnosis,

and tracking progress related to postural irregularities [8, 9]. Developed techniques will enable an increased ability to identify postural irregularities and respond to trends and customized plans based on the individual needs of different patients [3]. Furthermore, posture analysis can also contribute to improving treatment outcomes for patients by identifying potential imbalances at an early stage and enabling practitioners to evaluate treatment strategies and make adjustments accordingly [1, 4, 5].

In this scenario, the current project intends to complement this information gap in the available research on posture analysis systems to automatically identify anatomical points and estimate posture deviation [7, 10]. The system integrates computer vision technology and clinical research information to contribute objective information regarding diagnoses and treatments [11, 12]. Another intention expressed in this project is to complement a limitation presented by former systems to consider only relevant views related to posture [3]. This solution will complement the skills of healthcare specialists by providing credible information for more accurate diagnostics and rehabilitation [8, 2].

Recent advances in computer vision and machine learning have offered promising alternatives for non-invasive posture assessment [13, 14]. In particular, two-dimensional pose estimation methods [11, 12, 15] offer a low-cost and fast way to assess human posture by automatically selecting keypoint values from images. OpenPose, HRNet, MoveNet, and MediaPipe pose estimation libraries [7, 10] offer a robust approach to detect prominent joints and pose points in images [3]. By leveraging machine learning models trained on large datasets of human poses, 2D pose estimation opens new avenues for digital healthcare applications and real-time posture monitoring [6, 5].

Despite these advances, current automated pose estimation methods face several challenges when applied to real-world clinical data [8, 16]. Lighting conditions, clothing types, body shape, camera angles, and occlusions may affect accuracy [17]. Minor inaccuracies in the identification of important key points such as shoulders, hips, and the C7 vertebra may greatly affect the accuracy of angle identification or postural misalignment [3]. This highlights the importance of preprocessing, validation, and annotation credibility in automated posture analysis systems [16, 17].

This project utilized a data set reviewed by Dr. Mahmoud Ibrahim, consisting of clinical images of patients diagnosed with postural abnormalities. The images were taken from various standing positions. To simplify the analysis and focus on clinically relevant perspectives, this study concentrates on the front and back views, which provide the most information about spinal alignment, shoulder symmetry, and trunk balance [3]. A subset of images was annotated to create a ground truth dataset for evaluating automated keypoint detection [11, 12].

The approach adopted in this work integrates preprocessing, automated keypoint detection using the MediaPipe Pose model pretrained on 33 anatomical landmarks, and the derivation of custom clinical points including the mid-hip, torso center, C7, and auxiliary points aligned with the torso center [7, 10, 3]. These points enable the calculation of deviation angle, leaning direction, and postural stability, with validation against manually annotated ground truth [6].

In conclusion, this thesis continues work on improving 2D image-based posture estimation by combining automated keypoint detection, geometric computations, and clinical verification [7, 3]. It addresses key challenges related to accuracy, anatomical variability, and interpretability, and demonstrates how pose estimation techniques can be optimized for medical posture assessment and future applications in rehabilitation and ergonomic analysis [6, 5, 8].

Chapter 3

Background and Literature Review

3.1 Background

Human posture recognition is considered a fundamental problem in both computer vision and human-computer interaction since it addresses most of the applications in ergonomics, health monitoring, rehabilitation, and human-machine interfacing [9]. Traditional analysis of posture has been performed using specialist sensors or manual annotation, although recent advances in machine learning have accelerated the development of automatic posture understanding from both visual and sensor data [13].

These postural problems can present themselves in a variety of forms such as shoulder misalignment, lateral bending, spinal curvature, or leaning on the body as a whole. Among the postural disorders that can be identified is that of scoliosis, which can be considered a lateral deviation in the curvature of the spine, whereby the patient's body is leaning either on the right or on the left [1, 3]. Scoliosis is a very common spinal deformity that usually attacks adolescents and can progressively change the symmetry and balance of the trunk [1, 3]. The intensity of this posture can range in severity such that functional problems in mobility can develop that would necessitate clinical intervention in order to forestall the exacerbation [3]. Early recognition of such imbalances, including deviations in trunk position and asymmetry, is important for effective screening, monitoring, and intervention [1, 3]. In this particular research, an examination would specifically be on issues where the patient leans either on the right or on the left, which often requires specialist assessment to determine clinical significance and possible underlying scoliosis [2].

The approach which has shown promise is the application of pose estimation networks to directly solve the task of identifying keypoints in the human body from RGB images [7]. Such pose estimation networks enable the pose information which is required for higher-level analysis, such as the analysis of posture, to be extracted from the input. Amongst the modern pose estimation techniques, the technique which has shown to be efficient for real-time processing of pose estimation, enabling real-time posture analysis, is the BlazePose technique [7].

3.2 Literature Review

3.2.1 Machine Learning Pose Estimation Studies

A review paper compared multiple MLPems like OpenPose, HRNet, AlphaPose and MoveNet. It used public datasets like COCO, Human3.6M and MuPoTs-3D datasets. However, this research paper is comparative and doesn't include any implementation. Another limitation includes limited standardisation and it uses few longitudinal studies. In addition to this, it used tools like PyTorch and TensorFlow as well as preprocessing steps like normalization and keypoint alignment. In this paper to evaluate the models, it used cross dataset analysis. From this paper we can conclude that OpenPose is widely used while HRNet and MoveNet are more accurate [6].

Another research paper also used MLPems for posture automation. This study extracts keypoints and used OpenPose and Deep Learning models with RULA scoring [4]. In order to evaluate the model this paper used accuracy as an evaluation metric. However, this paper faces problems in different lighting and camera positions [4]. The result of this study is an OpenPose that can automate the posture and it was accepted by experts and specialists.

Another study used AlphaPose and MediaPipe 3D and MediaPipe 2D in order to be able detect low-cost solutions for identifying motor impairments. This paper used videos so we used frames extractions and to be able to evaluate the model, it uses AVOVA and Correlation [8]. From this paper, we can conclude that MediaPipe 3D has the best correlation. However, this paper contains occlusion and data about small people only [8].

A research study uses MoveNet and iPhone 13 in order to compare pose parameters against X rays angles. This paper uses iPhone 13 because we work on clinical images using a smartphone and mobile devices. It uses preprocessing steps like cropping and pose extraction[3]. It also uses Pearson Correlation as an evaluation metric. From this paper, we can conclude from the results that there is a high correlation between clinical pictures and X ray angles. However, one of the limitation of this paper is that there is BMI bias and the dataset is small[3].

3.2.2 Machine Learning and Deep Learning Approaches for Posture Analysis

This research paper includes videos about skeleton data and its aim is to assess WMSD risks. It used tools like MATLAB and Kinect SDK. This study uses preprocessing steps like depth calibration filtering noise. This study uses R-CNN and Fuzzy Logic and to evaluate these architectures it uses manual and auto RULA. Result for this paper is that each model outputs automated and consistent score for the posture (accuracy) [18]. Although this paper proves the accuracy of a posture analytical system, it is still a prototype with many limitations before it can be used in the field. The study already

mentions many of these limitations, and it should be noted that these areas will be targeted by the authors in their next line of research. There is more methodological innovation here than application, in terms of proving that computer vision and fuzzy-logic techniques can match expert raters on RULA in controlled settings, but not for scaled implementation on-site [18].

This paper presented a significant advance in automated biomechanical screening with the development of a video-based system for gait analysis specifically tailored to spinal deformity assessment[16]. Their approach addresses a major technical challenge of real-time depth completion by proposing a new two-branch CNN architecture enhanced by residual learning. Given a substantial database of 20,000 RGB-Depth pairs for training, their model takes input that is both denoised and normalized to predict the accurate depth map from standard video. Strong performance was established, showing a robust 30 FPS suitable for clinical real-time use while measured against standard error metrics such as RMSE. Implemented in PyTorch, this deep learning approach can be seen to represent a shift from traditional marker-based motion capture to accessible, vision-based analysis. The authors still mentioned several of the relatively persistent limitations associated with the optical systems themselves: sensitivity to visual noise and degraded performance in ill-lit environments. This work represents an increasing tendency to use efficient designs of CNNs to enable sophisticated biomechanical assessment outside controlled laboratory conditions[16].

A research paper done by Lee et al. in 2024 proposed a deep learning pipeline for video-based detection of FHP [19]. The method first estimates 2D human pose using Detectron2, followed by the reconstruction of a 3D skeletal sequence using VideoPose3D. Lastly, a graph convolutional network (GCN) classifies the 3D pose data into FHP or normal. The model was trained on a data split of 70/15/15 and obtained a classification accuracy of 78.3. The authors further pointed out the key limitation: the system classified but did not output the continuous CVA angle, which is the clinical standard for measurement. Furthermore, performance can be constrained by a small dataset. [19].

One of the biggest issues when evaluating the spine is that reliable 3D morphometric information cannot be gathered non-invasively without exposing patients to ionized radiation through X-ray or CT scans. To this end, there has been a recent investigation into vision technology methods that could offer a radiation-free alternative. With this objective in mind, a relevant study was that of Li, Chen, Zhao, and Zhang (2023), who made a valuable input within this theme through their Dual-Feature Extractor framework that aimed at reconstructing back depth images and spinal morphology solely from monochrome 2D images [17]. In this regard, they developed a specific architecture through their GAMA-Net framework that features a U-Net structure, symbolizing a specific response within this non-explored research field that shifted focus away from generic depth estimation networks. This study managed impressive results, including 97 percent segmentation accuracy and a low RMSE measure of just below 0.06, thus confirming that feasible, objective spinal parameters could be estimated from a single 2D image. Nevertheless, this study rests within a promising research borderland that is both radiation-free spinal evaluation and highly advanced computer vision applications, thus sandwiched between

classical biomaterials analysis [17].

Gupta, Sharma and Singh covered the comparison between traditional machine learning methods and models like (SVM and KNN) and deep learning models like CNN and DNN. This paper is a review-based. The results obtained from this paper is that DNNs are much better compared to SVM and KNN models when it comes to dealing with nonlinear posture because the models are able to extract hierarchical features from the data much better than the existing approaches (SVM and KNN). [5].

A research paper done by Khana et al. in 2022 offered a comprehensive review of both 2D and 3D human pose estimation methods in deep learning approaches, targeting the advances and challenges in the methods. The research work compared regression methods, heatmap methods, and the strategies of converting 2D to 3D using public benchmarked datasets such as COCO, Human3.6M, and MuPoTs-3D. The research work proposed the use of image normalization and keypoint alignment for greater precision in estimating poses by applying preprocessing methods. The methods employed for evaluating the work include quantitative Mean Per Joint Position Error and Percentage of Correct Keypoints [14]. The research work proved the effectiveness of using deep learning methods in accurately estimating poses compared to the previous methods, with the multiple view strategies showing greater precision compared to the single view strategies. Even though the results demonstrated the effectiveness of the deep learning methods, the research work argued that the methods still face problems of higher computational complexity and dataset bias. The research work was conducted using the PyTorch platform, ensuring the effectiveness and applicability of the deep learning concept [14].

A research paper done by Cheung et al. in 2020 worked on the feasibility study to use two-dimensional photogrammetry in a non-invasive technique for patient screening and monitoring related to adolescent idiopathic scoliosis. This work stresses a method to evaluate patient posture and spinal alignment via image processing approaches, with a goal to create a non-ionizing and safe alternative to frequent radiographic examinations. This study utilized edge detection and alignment to pick out image features concerning patient posture, thereby applying manual photogrammetry analysis for statistical verification. This method applied two different parameters, namely Intraclass Correlation Coefficient (ICC) and Pearson Correlation, to measure the consistency level between photogrammetry values and validated values obtained via radiographic methods [1]. The result shows excellent consistency values[1]. This study emphasizes the viability and feasibility of applying 2D photogrammetry in a safe, non-ionizing, and inexpensive method for conducting posture analysis. This study, however, describes some limitations concerning the angle limitations imposed upon image perspectives. This study applied image processing software named Photoshop, emphasizing the feasibility, acceptability, and concrete constraints requiring enhanced study in developing an intelligent multi-view alignment system [1].

Even though the estimation of pose models has been very effective in terms of joint detection accuracy, it is very important to comprehend their limitations and performance capabilities in real-world applications. The assessment of pose estimation methods is highlighted in thorough surveys of estimation techniques, where measures such as Percentage

of Correct Keypoints (PCK) play important roles in comprehending their performance capabilities [13].

However, beyond the basic task of keypoint detection, studies on posture recognition have tended to emphasize the classification of posture quality; for example, distinguishing between aligned and misaligned postures [2]. Recent studies such as "Decoding Posture: Distinguishing Between Good and Bad Body Alignment" have shown the need to evaluate the correctness of posture, beyond localizing joints in a subject's body.

Another area that research conducted simultaneously for posture recognition in specialized fields such as privacy-preserving thermal imaging for posture recognition, where deep learning can classify overall posture categories while preserving identity [9]. While these topics vary greatly from one area to the next with regard to the nature of the work performed, they each verify the overall need for posture evaluation with regard to accuracy and application.

Despite the above progress, there is a challenge in systematically assessing the correctness of pose estimators against annotated keypoints and relating them to posture assessment metrics [13]. A common characteristic of posture analysis systems is the assumption that keypoints are sufficiently accurate without a complete assessment against annotated keypoints. This often may lead to inaccurate posture assessment if keypoints are not near their true positions [13].

In this research, the issue has been filled by integrating pose estimation with thorough validation against labeled keypoints with the usage of metrics like Percentage of Correct Keypoints (PCK), Mean Absolute Error (MAE), and analysis of the angle of divergence [7, 13]. The pose analysis has been performed by utilizing the pre-trained MediaPipe Pose solution to detect skeletal keypoints from an RGB image and calculate secondary keypoints essential to the analysis of posture analysis like C7, Mid-hip, and the center of the torso [7].

The results obtained from the pose analysis are then validated with the ground truth by using LabelMe to assess the precision of the keypoints identified as well as the authenticity of the posture divergence measurements obtained [2].

Chapter 4

Methodology

This chapter describes the methodology followed to design, implement, and evaluate the proposed posture analysis system. The overall workflow includes data acquisition, preprocessing, pose estimation using a deep learning model, computation of posture deviation metrics, and result visualization through reports and dashboards. Two different implementation strategies were investigated in order to assess their effectiveness: separating front and back views, and merging front and back views into a unified training process.

4.1 System Overview

The proposed system follows a structured pipeline designed to ensure accurate posture analysis and reproducibility. The flow of work begins with image acquisition and ends with evaluating the posture evaluation and reporting it.

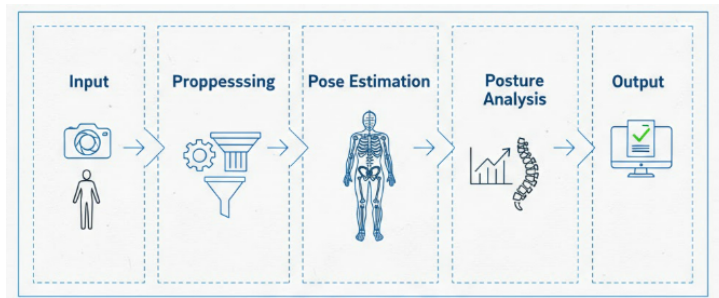


Figure 4.1: Proposed Human Posture Estimation Architecture

The architecture has five major components: (a) Input, where images and/or videos are fed as an input to the system, (b) Preprocessing, where normalization of data and preprocessing take place, (c) Pose Estimation, where detection of human posture points happens, (d) Posture Analysis, where posture analysis is carried out, and (e) Output, where the final posture analysis and metrics.

To begin with, the RGB images of the patients are taken from both the front and the back. Next, these images are preprocessed in order to increase data consistency and remove noisy data [16, 17]. The preprocessed images are passed to a pose estimation model based on MediaPipe Pose, which detects anatomical landmarks [7, 10]. During the posture analysis, from these landmarks, posture-related measurements such as the deviation angle are computed [3]. Finally, the results are presented in the form of reports and dashboards that summarize posture quality, track changes over time, and provide recommendations [8, 9].

4.2 Dataset Description

The dataset used in this study consists of posture images collected from multiple patients and from multiple clinics across Egypt under the supervision of Dr. Mahmoud Ibrahim. For each patient, images were captured from both front and back views to allow a comprehensive assessment of body alignment.

The dataset includes a total of 6,404 images. For each patient, a set of front and back images was collected as in figures 5.2 and 5.3, and then they were manually annotated and exported in JSON format under the supervision of specialists and experts. These images were manually annotated to define key anatomical landmarks required for posture analysis using a tool named LabelMe [20]. These annotations serve as ground truth data for training and evaluating the pose estimation model.



Figure 4.2: Sample front image from the dataset.

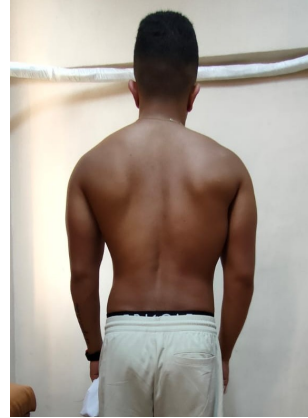


Figure 4.3: Sample back image from the dataset.

4.3 Preprocessing

Before moving on to the pose estimation stage, a series of preprocessing activities had been done on all the images. The activities were important in ensuring that the dataset was standardized.

4.3.1 Background Removal

Removal of the background was used to filter irrelevant visual information to direct attention to the human body [7]. This process enhances accuracy while detecting landmarks [10].

4.3.2 Normalization

This also normalizes images to have a standard value of pixel intensity. This helps to stabilize the learning process and generalize better in various images [16, 17].

4.3.3 Cropping

Cropping was applied to center the subject within the image frame [7]. Dropping all the unwanted areas around the subject allows the model to focus better on the required regions of the body [10].

4.3.4 Reflection "Mirroring Images"

Mirroring was applied as a data augmentation technique to increase dataset diversity [16]. This step helps the model learn posture patterns symmetrically and improves robustness to left-right variations[17].

These preprocessing steps were applied consistently to both front and back view images.

4.4 Pose Estimation

There were two different approaches used in order apply the pose estimation. Both approaches were based on the MediaPipe architecture.

4.4.1 MediaPipe Pose Model

MediaPipe Pose is a deep learning-based solution for human pose estimation and tracking developed by Google, designed for efficient processing of both images and video. This model provides 33 pre-defined anatomical landmarks in both 2D and 3D, which correspond to major joints and body parts such as shoulders, hips, and spine-related landmarks [7, 10]. MediaPipe Pose has also gained massive interest in posture analysis systems, including medical domains [13, 6, 3], due to its balanced performance regarding estimation accuracy and computational efficiency.

In this work, the MediaPipe Pose model has been adopted to extract elaborated body landmarks from RGB images serving as the basic building blocks to calculate posture-related measurements [7, 10]. Although the pre-defined landmarks are vital for their joint-level information, further derived keypoints that are posture-specific have been obtained to facilitate more detailed evaluations of alignment of the body. These derived keypoints include the center of hips, center of torso, an approximated C7 point, and two supporting reference points referred to as point a and point b; all geometrically computed from the primary MediaPipe landmarks.

The calculation of these extra keypoints is based on geometric and photogrammetric relations that are well established in posture analysis literature [1]. Given that all of the above posture metrics rely on multiple, anatomically related landmarks rather than using individual joint positions, posture metrics may be computed from a more global view of body alignment. In this way, local fluctuations in individual landmarks are less influential, enhancing robustness and stability in measurement of postural deviation.

With this, the posture analysis is found to have a reliable backbone through the integration of the MediaPipe Pose model, while the deliberate selection and computation of anatomically meaningful keypoints allow the proposed system [10] to estimate posture deviations accurately.

4.4.2 Implementation Approaches for View-Based Posture Analysis

There are three different experiments used while keeping an identical model architecture, training and configuration across all approaches. The only two factors that change between these experiments is the strategy used to organize the dataset at the patient level and whether the annotations is automatic or manual. The first experiment was to separate front and back images while the other experiment was to merge front and back images. In the first approach "front and back specific training", automatic annotations were used while the other two approaches "combination of front and back specific training" and "patient-specific approach" were implemented using manual annotations. To use automatic annotations, a loop iterates over all patients to automatically annotate each patient using keypoints derived from MediaPipe, while other keypoints are calculated geometrically. After drawing the points on the patient's body, they are saved as JSON files. This design ensures that the difference in each experiment's performance relies on the dataset organization and the annotations and not model-related variations. Using one pose estimation model enabled a differentiation of variations in posture estimation results based on data management methods, not models.

Front and Back Specific Training

In the first implementation strategy, front-view and back-view images were treated as two separate datasets. The motivation behind this approach was to allow the model to

specialize in view-specific posture characteristics [10]. Two separate models were used in this approach to train the back and front images. In this method, for every dataset formed, a batch of front images was added to the front dataset, and a batch of back images were added to the back dataset.

After preprocessing, back and front images were separately and automatically annotated using JSON files for each patient [11]. MediaPipe Pose model was then defined and trained using these annotations [7, 10]. The training process followed a transfer learning strategy. The first experiment is based on automatic annotations, and it was used for efficiency; allowing rapid creation of dataset and preliminary evaluation.

Combination of Front and Back Images Training

The second implementation strategy aimed to improve performance by combining front and back views into a single dataset [8]. The motivation for this approach was to increase the number of training samples and allow the model to learn posture characteristics across multiple viewpoints [10]. In this approach, we used a combination of front and back images to train only one model.

The same preprocessing steps described earlier were applied to all images [16, 17]. Each patient's data was processed and stored in a dedicated folder. Manual annotation was performed for all images, and the annotations were saved as JSON files [11].

Patient-Specific Approach

The MediaPipe Pose model was trained using the combined dataset, following the same transfer learning strategy used in the first method [7, 10]. The dataset here consists patient-specific folders where each patient has their own front and back images.

Particular attention was given to the upper body landmarks, since the upper body is more dependent on the landmarks in the torso area, which has additional relevance to the alignment of the posture. The landmarks on the shoulders and hips were given special prominence since the position of these two landmarks relative to each other contains key information on the leaning of the body.

To evaluate each approach implemented, the dataset size was expanded by 1000 images to analyse the model's performance. The first and second approaches were tested using PCK, MAE and MSE. In contrast, the patient-specific approach was tested using PCK, MSE and MAE to measure the accuracy and correctness of the detected keypoints on the patient's body.

4.5 Posture Analysis

4.5.1 Angle of Deviation Calculation

The deviation in posture was measured by finding the angle that was subtended by a vertical the reference line to the point connecting the C7 vertebra point to the body

center [2, 3]. This angle functions as a geometric marker for lateral alignment of the body and represents how well the patient's posture varies from the erect posture [4].

The upright line is known as the standard alignment, while the line connecting the C7 vertebra and body center accurately represents the orientation of the upper body [3]. This angle difference between the two lines leads to a distinct measure for posture deviation [2]. A negative or positive value shows the nature of leaning, such that it can be distinguished between left-leaning posture and right-leaning posture [3].

Posture deviation was calculated by drawing a virtual line connecting the C7 vertebra point and the torso center [3]. From this line, the deviation angle was computed for each image [2]. For each patient, the average and maximum deviation angles were calculated [3].

Using the computed angle [2, 3]:

- Small angles mean good alignment of the posture and least amount of deviation from vertical reference [2].
- Larger angles reflect increased posture deviation, showing marked lateral forward leaning of the upper body [3].

The images in Fig. 4.4 show examples of where each anatomical keypoint is located on the front and back views of a patient.

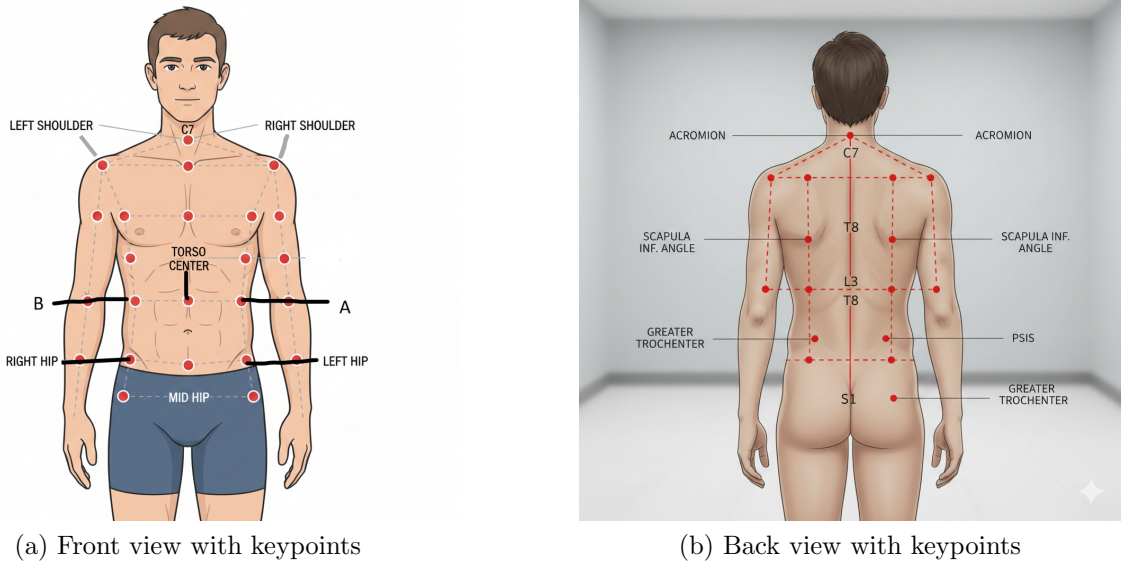


Figure 4.4: Examples of annotated keypoints in the dataset.

For each patient, deviation angles were determined for all applicable images and assessed together in order to make an evaluation of overall posture behavior [8, 3]. This will allow for easier evaluation of posture due to between different subjects as well as between multiple observations on the same subject [9].

Based on these values, a report was generated containing:

- Deviation angle for each front and back image [8]
- Average deviation angle [3]
- Maximum deviation angle [3]
- Overall leaning direction [3]

Each patient's posture was finally categorized based on direction as well as on severity of the deviation, offering a brief but telling summary of posture alignment which can be used again in further evaluation [3, 5].

4.5.2 Evaluation Metrics

The performance of the pose estimation model is measured through the Percentage of Correct Keypoints on the target image [13, 6]. Correct Keypoints (PCK) metric. PCK has been a widely used evaluation metric in human pose estimation problems and is used for evaluating the accuracy of the keypoints with corresponding ground truth annotations available within a certain predefined threshold [13, 5]. MSE and MAE were also used in this research[5, 13]. MSE and MAE were used to provide a pixel-wise distance between predicted and actual keypoints. However they do not indicate whether a keypoint is anatomically correct or not [3, 8]. Using all these metrics all together provide a more complete evaluation and differentiates between low numerical error and true anatomical accuracy.

This gives an intuitive and easy-to-understand measure of the keypoint localization accuracy [3]. There's no adjustment to precision in terms of determining the measure of correctly recognized landmarks [8]. Posture analysis depends entirely on the accurate location of keypoints in the skeleton, the application of PCK makes it possible for the reliability of the pose estimation step should be evaluated in a consistent and objective manner [7, 10].

It was chosen as it is a widely applied technique in literature and appropriate to measure models that work with annotated landmark points [13, 6]. It emphasizes the accuracy of individual keypoint predictions, the measure provides information on the model's capability to capture human body structure correctly [3].

In evaluating the performance of pose estimation using PCK, the succeeding posture deviation calculations are founded on reliable landmark identifications [8, 3]. This further enhances confidence in the validity of the reported posture measurements and facilitates meaningful interpretation of the system's final outputs [2, 5].

4.5.3 Report and Dashboard Generation

The final process in the methodology dealt with the creation of reports and dashboards to enable the assessment and analysis of the posture. This process is responsible for the transformation of the numeric results from the posture analysis process to a format that is easy to interpret.

The reports contain structured numerical data on the measurements for posture deviation, as well as angles for deviation that are determined for patients. A quick summary and review for patients are achievable because data is presented in an organized fashion and manner.

Report includes the needed data like:

- name of patient
- deviation angle
- average deviation angle
- maximum deviation angle
- total images analyzed
- leaning direction
- name of the image (front and back images)

Besides the reporting, dashboards were created to display the data on posture in a visual format. The dashboards facilitate the monitoring of the changes in posture over time in an intuitive manner. Visualization of the data allows for easier interpretation but enable users to judge the posture status remotely without needing expert knowledge.

The dashboards are also equipped with suggestions on how to improve posture based on the analysis results. These suggestions enable the medical practitioners or users to track the progress made in posture analysis and detect any possible posture problems in a convenient manner. These suggestions are given based on the maximum deviation angle. The analysis involves the use of numeric summaries with visual feedback, the report interface and dashboard part of the solution improve the usability of the proposed solution in general.

Dashboard includes the needed data such as:

- name of patient
- deviation angle
- average deviation angle

- maximum deviation angle
- total images analyzed
- date
- leaning direction
- recommendations on how to improve the posture

Chapter 5

Results

5.1 Front and Back Specific Training

In the first experiment, two separate and independent datasets of images of patients from the front and from the back were developed. The model was developed using the MediaPipe algorithm and automatic JSON annotations to mark the essential points of the human anatomy to be identified. The motive behind this approach was to evaluate each view separately in order to analyze the model's performance for each view.

In the quantitative assessment, Mean Squared Error (MSE) and Mean Absolute Error (MAE) were used as metrics for evaluation [13, 6]. In the case of back-view images, the proposed approach obtained an MSE of 0.0018 and an MAE of 0.0451, while for front-view images, the model demonstrated low errors. For the front view, the MSE was 0.00085 and the MAE was 0.0298. Metric like PCK was also used in this approach. The results of the PCK are as follows for the front view: left shoulder = 0.82, right shoulder = 0.80, C7 = 0.52, torso center = 0.71, A = 0.55, B = 0.54, right hip = 0.56, left hip = 0.58, mid hip = 0.60.

The results of the PCK are as follows for the back view: left shoulder = 0.79, right shoulder = 0.78, C7 = 0.50, torso center = 0.70, A = 0.52, B = 0.51, right hip = 0.53, left hip = 0.55, mid hip = 0.57.

Although these error values are small, qualitative analysis did show the existence of discrepancies with regard to the location of the keypoints. The model performed errors especially in anatomical regions such as the hips and shoulders. From this observation, it can be concluded that MSE and MAE values do not capture the correctness of the keypoints annotated on the patient's body.

These disadvantages initiated a second approach, which was based on manual annotation and the merging of datasets. The first approach relied on automatic annotation for efficiency; it enables rapid dataset creation and preliminary evaluation. However, this automatic approach does not have enough detail regarding anatomy, which again motivated performing the switch to manual annotation.

Table 5.1: Result of Separating Front and Back Images With Automatic Annotations

View Type	MSE	MAE
Front Images	0.00085	0.0298
Back View Images	0.0018	0.0451

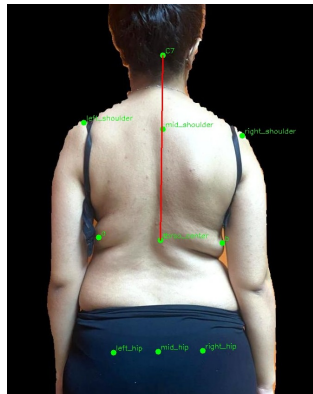


Figure 5.1: Annotated back-view image.

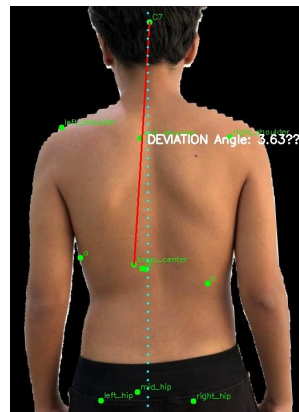


Figure 5.2: Model output for back-view images after training.

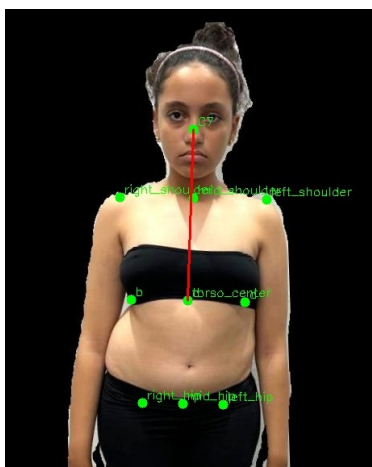


Figure 5.3: Annotated front-view image.

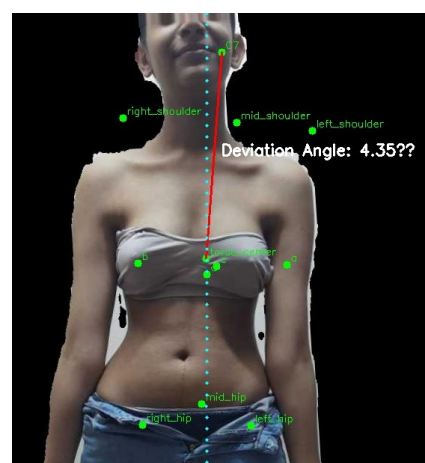


Figure 5.4: Model output for front-view images after training.

5.2 Combination of Front Views and Back Images Training

In the second experiment, images of the front and back views of the patients were merged into a common dataset. This was done to increase the pool of training instances.

Next, the trained model was tested on the merged dataset and acquired a Mean Squared Error of 0.0109 and Mean Absolute Error of 0.0944. Although these error values were somewhat higher than those obtained in the first experiment, the manually annotated and combined dataset resulted in more consistent and anatomically meaningful keypoint predictions. PCK was also used in this approach and the results are as follows: Left shoulder = 0.93, Right shoulder: 0.92, Left hip: 0.68, Right hip: 0.66, Mid hip: 0.67, Torso center: 0.89, Point A: 0.74, Point B: 0.73, C7: 0.55.

Evaluation in this study was carried out using the Percentage of Correct Keypoints (PCK), a widely accepted metric for assessing spatial accuracy in pose estimation research [13, 5]. While MSE and MAE are commonly used as loss functions during training, they do not directly reflect the anatomical correctness of keypoint placement in comparison to PCK. These findings motivated the transition to a pretrained pose estimation model and the adoption of keypoint-specific evaluation metrics.

Table 5.2: Results of Merging Front and Back Views with Manual Annotation

Metric	Value
Mean Squared Error (MSE)	0.0109
Mean Absolute Error (MAE)	0.0944

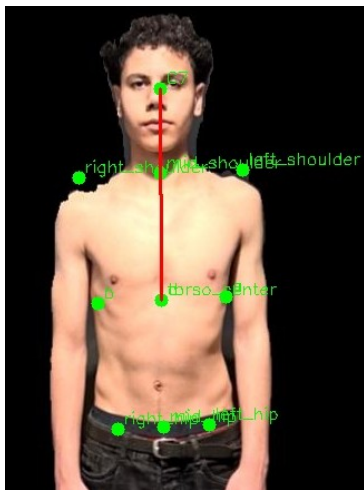


Figure 5.5: Annotated front-view image.

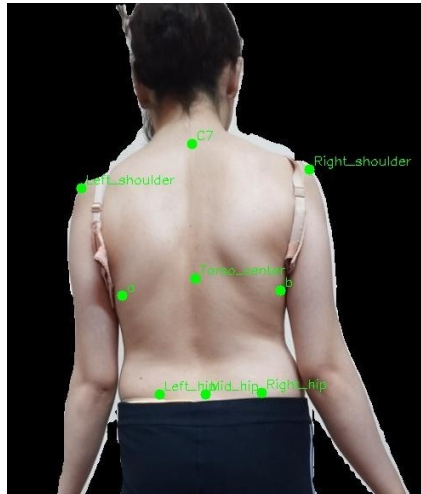


Figure 5.6: Annotated back-view image.

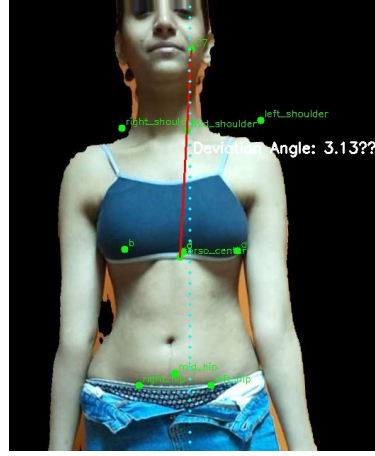


Figure 5.7: Model output for front-view images after merging front and back views.

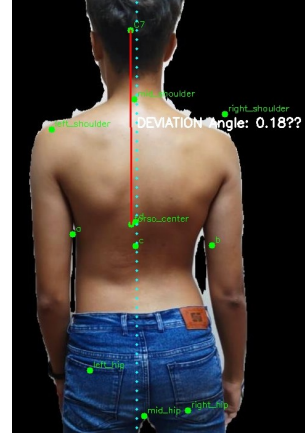


Figure 5.8: Model output for back-view images after merging front and back views.

5.3 Patient-Specific Approach

In the last experiment, the MediaPipe Pose model, which has a pre-trained 33 marks of the human body, was utilized [10]. Rather than fine-tuning all 33 landmarks, the model was fine-tuned for the assigned marks of the shoulders and hip. Other posture-related marks, namely points C7, mid-hip, torso, and points A and B, were calculated based on geometric equations. To assess its accuracy, the Percentage of Correct Keypoints (PCK) method, comparing the reckoned keypoints to those marked by human annotators, was utilized [13, 5]. Results of the acquired PCK are as follows: Left shoulder = 0.927, Right shoulder = 0.917, Left hip = 0.660, Right hip = 0.658, A = 0.776, B = 0.777, mid hip = 0.710, C7 = 0.937, center of torso = 0.884. MSE is 0.0048 and MAE is 0.0396 respectively. In contrast to the previous two approaches, this approach shows improvement in annotating the keypoints on the body using the PCK metric.

These experiments clearly show a high level of accuracy on upper-body points, and C7 and shoulders are particularly relevant, since these are fundamental for analyzing posture. The lower PCK values on hip-related points indicate higher variability because of clothing, occlusions, and viewpoint variations.

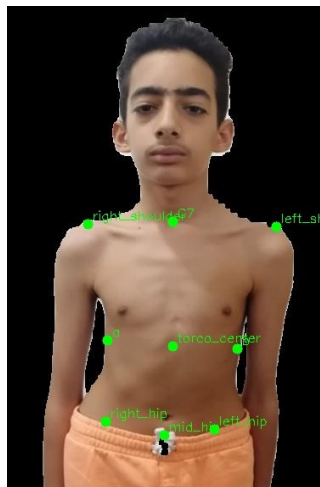


Figure 5.9: Example of an annotated front-view image using MediaPipe.

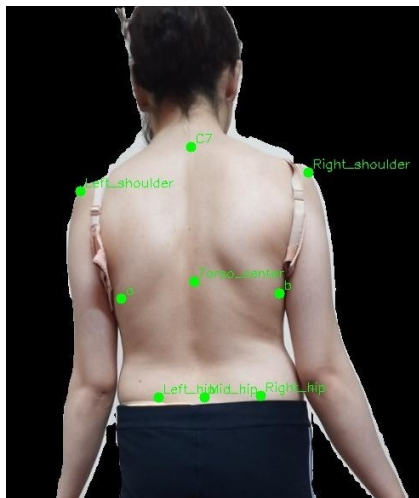


Figure 5.10: Example of an annotated back-view image using MediaPipe.

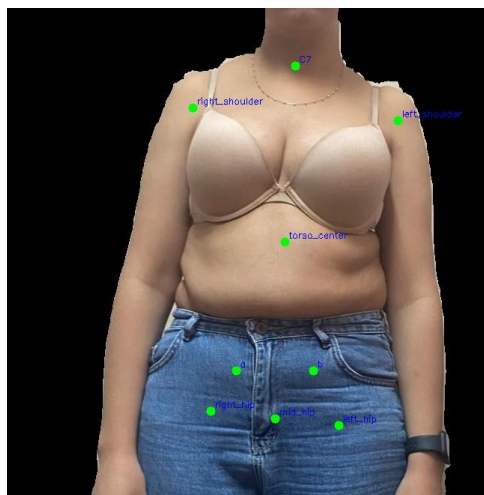


Figure 5.11: Output of MediaPipe pose annotation for a front-view image.

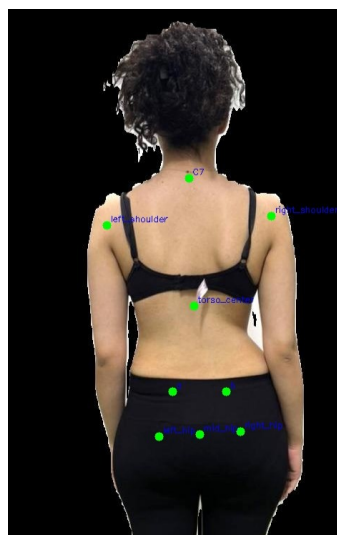


Figure 5.12: Output of MediaPipe pose annotation for a back-view image.

Based on the accurately localized keypoints, the angle of deviation was computed for each patient to determine left or right postural leaning. Patients were classified into multiple cases and categories including well-aligned, mild deviation, moderate deviation, and severe deviation.

5.3.1 Report and Dashboard

A detailed report and an interactive dashboard were generated to present deviation angles, leaning direction, average and maximum deviation values, and temporal posture tracking across multiple visits. The dashboard additionally provides personalized posture correction recommendations based on deviation severity.

Patient Name: Jana Mahmoud

Summary Metrics

Total images analyzed: 12
Average deviation: -1.05°
Maximum deviation: 0.57°
Overall leaning direction: Left

Daily Posture Metrics

Date	Total Images	Avg Deviation (°)	Max Deviation (°)
2023-08-26	0	0.00	0.00
2023-09-13	2	0.20	0.43
2023-10-11	1	-3.04	-3.04
2023-11-06	0	0.00	0.00
2023-12-11	2	-2.04	-1.39
2024-05-11	2	-2.81	-0.77
2024-11-02	4	-1.30	0.21
2025-01-30	1	0.57	0.57

Recommendations

• Posture is well aligned. Maintain good habits.

POSTURE ANALYSIS REPORT – AHMED TAREK

Total images analyzed: 14
Average deviation: 160.91°
Maximum deviation: 165.10°
Overall leaning direction: Right

Deviation Angle Per Image

Image Name	Deviation Angle (°)
pic1_19-8-2025.jpg	162.28
pic2_19-8-2025.jpg	162.49
pic6_12-3-2025.jpg	162.75
pic7_12-3-2025.jpg	160.79
pic8_15-2-2025.jpg	163.43
pic9_15-2-2025.jpg	159.73
pic10_15-2-2025.jpg	162.13
pic11_16-11-2024.jpg	159.95
pic12_16-11-2024.jpg	156.37
pic13_17-8-2024.jpg	163.06
pic14_17-8-2024.jpg	161.86
pic4_2-2-2024.jpg	165.10
pic3_2-2-2024.jpg	153.78
pic5_4-1-2025.jpg	158.99

Figure 5.13: Example of the dashboard for a patient.

Figure 5.14: Example of the report for a patient.

Table 5.3: Comparison of All Implemented Methods

Method Type	Type of Annotation	Evaluation Metric	Advantage and Disadvantage
Separate Views	Automatic	front PCK = 0.631 back PCK = 0.605 MSE = 0.00085 MAE = 0.0298	Advantage: View-specific posture learning Disadvantage: Low anatomical accuracy
Combined Front and Back Views	Manual Annotation	PCK = 0.752 MSE = 0.0109 MAE = 0.0944	Advantage: Improved generalization across views Disadvantage: MSE and MAE are unsuitable for posture analysis
Patient-Specific	Manual Annotation	PCK = 0.805 MSE = 0.0048 MAE = 0.0396	Advantage: High clinical interpretability Disadvantage: Hip variability

Chapter 6

Discussion

The proposed posture analysis system successfully integrates both predefined anatomical landmarks from the MediaPipe Pose model and additional custom landmarks required for clinical posture assessment. By combining standard keypoints such as shoulders and hips with derived points including C7, mid-hip, torso center, and the auxiliary points A and B, the system provides a comprehensive representation of upper body alignment for both front and back views.

The evaluation results demonstrate that the system achieves high accuracy for upper-body landmarks, particularly the shoulders and the C7 point, as reflected by their high PCK values. This indicates that MediaPipe Pose is highly reliable in detecting major anatomical joints that are clearly visible in RGB images. However, lower PCK scores were observed for the hip-related landmarks and the derived points A and B. This performance gap can be attributed to several factors, including partial occlusion of the hip region, variability in clothing, viewpoint differences, and the reliance on manual annotations for ground truth labeling.

In addition, the shoulder landmarks, although accurately detected, were observed to be slightly lower than their anatomically ideal positions in some cases. This inconsistency might affect the accuracy of posture estimation, especially in the healthcare application in which small angular differences could be considered important. It would be helpful to adjust the shoulder keypoints upwards in order to get more accurate values for deviation angle calculation and make the system more useful in health care.

The quality of annotations can also influence evaluation results significantly. Poor and inconsistent annotations of hips and generated points can directly impact PCK performance, which emphasizes the priority of refined annotations and expertise in label preparation of the dataset. These aspects are expected to greatly contribute towards improvement in keypoint detection accuracy and generated points, which are not directly predicted by the pre-trained model.

In summary, the outcomes prove the effectiveness of the combination of a pose estimation network with geometrical landmarks for posture analysis. Though the system

presented has a good performance level, further research should aim to improve annotation techniques and anatomical constraints to increase system robustness. In this matter, increased precision for posture diagnosis would bring significant advantages to the health-care field, which would benefit from improved resources for professionals to make correct diagnoses based on posture analysis information.

Chapter 7

Conclusion

This research aimed to develop a robust and non-invasive system for human posture analysis using 2D images, focusing on front and back views of patients. By integrating pose estimation techniques with careful preprocessing, manual annotation, and geometric derivation of additional keypoints, the study addressed key challenges in automated posture assessment, including the limited accuracy of pose detection models under clinical conditions.

The proposed system successfully detects critical anatomical landmarks, such as shoulders, hips, C7, mid-hip, torso center, and auxiliary points “A” and “B,” which are essential for assessing body alignment and deviation. By calculating deviation angles and analyzing leaning directions, the system provides quantitative metrics for evaluating postural misalignment. It also allows for temporal tracking of the patients’ posture through reports and dashboards generated from the system, whereby clinicians can track changes over time and give personalized recommendations on posture improvement.

Therein, it explored three different approaches to implementing: training for front and back views separately, data augmentation by combining front and back images, and a pre-trained MediaPipe Pose model combined with custom keypoints. The results indicate that while automated keypoint detection offers substantial efficiency, its accuracy improves significantly when combined with careful preprocessing and manual annotation. The use of Percentage of Correct Keypoints (PCK) as an evaluation metric demonstrated the effectiveness of the system in capturing patient posture accurately, although certain points, such as shoulders and hips, require further refinement to achieve optimal clinical utility.

The contributions of this research are multifold:

1. **Integration of Pose Estimation with Custom Landmarks:** By adding clinically relevant keypoints derived geometrically, the system enhances standard pose estimation frameworks to better capture postural deviations.

2. **Systematic Evaluation and Validation:** The comparison between automated and manual annotations, along with metrics like PCK, MSE, and MAE, ensures that the system's predictions are reliable and clinically meaningful.
3. **Practical Tools for Clinical Use:** The generated reports and dashboards provide a user-friendly interface for clinicians, enabling both immediate assessment and longitudinal tracking of patients' posture.

Despite these achievements, there are some downsides that still exist. The results of keypoint detection can still be influenced by occlusion, clothing, and large discrepancies in patients' physical sizes and shapes. Some of the calculated points, such as "A" and "B," still need fine-tuning so that accurate alignment with regard to the chest centre is possible. The current model only provides front and back views, and adding side and dynamics views might give a more complete evaluation of a person's posture.

Future research could remove these limitations by integrating more accurate pose estimation networks under difficult conditions, extending the data set to include more patients, and implementing information regarding dynamic posture changes from video data. Moreover, the integration of analysis from 2D images and data from sensors, such as Inertial Measurement Units, could increase robustness.

The conclusion of the research clearly shows the capability of a well-designed image processing posture analysis system in providing a non-invasive and accurate posture assessment of human beings through a 2D image. As a result, the developed solution made a prominent advancement in the field of posture analysis to offer a rehabilitation and preventative solution for human beings.

Appendix

Appendix A

Lists

1. MAE: Mean Absolute Error
2. MSE: Mean Squared Error
3. PCK: Percentage of Correct Keypoints
4. MLPEMS: Machine Learning Pose Estimation Models
5. 2D: 2 Dimensional
6. CNN: Convolutional Neural Network
7. RNN: Recurrent Neural Network
8. BMI: Body mass index
9. JSON: JavaScript Object Notation
10. RGB: Red, Green, Blue (color image)
11. HRNet: High Resolution Network
12. ANOVA: Analysis of Variance
13. C7: Seventh cervical vertebra
14. ICC: Intraclass Correlation Coefficient
15. PC: Pearson Correlation
16. 3D: 3 Dimensional
17. SVM: Support Vector Machine
18. KNN: K-Nearest Neighbour

19. **RMSE:** Root mean square error
20. **DNN:** Deep Neural Network
21. **MuPoTs-3D:** MuppetVision 3D
22. **Human3.6M:** 3.6 Million accurate 3D Human poses
23. **COCO:** Common Object In Context Dataset (large-scale object detection, segmentation, and captioning dataset)
24. **MPII:** Max Planck Institute for Informatics (Human Pose Dataset)
25. **CVA angle:** Costovertebral angle
26. **CT:** Computerized tomography
27. **GCN:** Graph Convolutional Network
28. **FHP:** Forward Head Detection
29. **FPS:** Frames Per Second
30. **RULA:** Rapid Upper Limb Assessment
31. **WMSD:** Work-related musculoskeletal disorders
32. **R-CNN:** Regions-based convolutional neural networks

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